$$\therefore P(X = x) = \frac{8C_x}{256}$$

P (getting atleast six heads)

$$= P(x \geq 6)$$

$$= P(x = 6) + P(x = 7) + P(x = 8)$$

$$= \frac{8C_6}{256} + \frac{8C_7}{256} + \frac{8C_8}{256}$$

$$= \frac{28}{256} + \frac{8}{256} + \frac{1}{256}$$

$$= \frac{37}{256}$$

Example 10.3

Ten coins are tossed simultaneously. Find the probability of getting (i) atleast seven heads (ii) exactly seven heads (iii) atmost seven heads.

Solution:

X denote the number of heads appear

$$P(X = x) = nC_x\, p^x\, q^{n-x},\ x = 0,1,2,\ldots n$$

Given: $p = P$ (head) $= \frac{1}{2}$ $q = 1 - p = 1 - \frac{1}{2} = \frac{1}{2}$ and $n = 10$

$$\therefore X \sim B\left(10, \frac{1}{2}\right)$$

$$= 10C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{10-x}$$

$$P(X = x) = \frac{10C_x}{1024}$$

(i) P (atleast seven heads)

$$P(X \geq 7) = P(x = 7) + P(x = 8) + P(x = 9) + P(x = 10)$$

$$= \frac{10C_7}{1024} + \frac{10C_8}{1024} + \frac{10C_9}{1024} \frac{10C_{10}}{1024}$$

$$= \frac{120}{1024} + \frac{45}{1024} + \frac{10}{1024} + \frac{1}{1024}$$

$$= \frac{176}{1024}$$

(ii) P (exactly 7 heads)

$$P(x=7) = \frac{10C_7}{1024} = \frac{120}{1024}$$